

Design An Experiment – Assignment Discussion

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Assignment Reminder

Many young investors misunderstand investment risk. You want to test whether presenting risk information visually improves their portfolio choices. In particular, you compare a standard text warning with a visual risk chart.

- Who are the participants?
- What is randomly assigned?
- What does the treatment change?
- What does the control group receive?
- What outcomes are measured?
- When are outcomes measured?

Compare a standard text warning with a visual risk chart.

The answers differed most in the experimental setting.

Overall Message

A good experiment section does not need to be complicated.

It needs to be explicit.

- What changes?
- What stays the same?
- What is measured?
- When is it measured?

Many imperfect answers contain strong research intuition.

Q1 Participants

Example

Young adults aged 18–30 who have limited to moderate investment experience.

What works

- Age range is specified
- Investment experience is considered
- Target population is not just “young people”

Example

300个年龄在20–35岁之间、拥有一定可投资资金但投资经验较少的年轻投资者.....

Revision question

- Finance students or ordinary investors?
- No experience, some experience, or active trading?
- Who should be excluded?

Q1 Clean Sentence

The study recruits young investors aged 18–30 with limited personal investment experience.

- Add recruitment source: university clubs, online communities, or brokerage users.
- Add exclusion criteria: professional finance workers or advanced finance training.
- Explain why this sample fits the research question.

Q2 Random Assignment

Example

平分成对照组和实验组，并确保两组在投资经验和教育背景等方面在统计上无显著差异。

Useful idea

- Two groups are imagined
- Baseline comparability matters

Revision point

- Say random assignment explicitly
- Do not randomize again until groups look good
- Use stratification or balance checks

Q2 Clean Sentence

Example

Participants are randomly assigned at the individual level into one of two groups.

Good writing move: name the unit of assignment.

Participants are randomly assigned at the individual level to either the treatment group or the control group.

Q3 Treatment

Example

The treatment changes only the way risk information is presented. Both groups see the same investment products, expected returns, fees, and basic descriptions.

The treatment changes only the way risk information is presented.

This is the cleanest version of the experiment.

Q3 Treatment

Example

在看到标准的文本风险警告之外，还会额外看到一张可视化风险图表.....

Useful idea

- Visual chart is clearly introduced
- The scenario is concrete

Design issue

- Text plus chart is not the same as chart vs text
- The treatment may add information
- Ask: what is the one clean difference?

Q4 Control Group

Example

The control group receives a standard written risk disclosure statement, identical in content to the information depicted in the visual chart, but presented as plain text without any graphical elements.

The control group receives the same underlying risk information in a standard text warning.

Control is not “no information.”

Q5 Outcomes

Example

The primary outcome is the percentage of the \$10,000 endowment allocated to the risky asset (0–100%).

Why this works

- Concrete and measurable
- Directly records portfolio choice
- Measured before market luck appears

Next improvement

- Lower risk is not always better
- Match choice to risk tolerance
- Add comprehension as secondary outcome

Q5 Outcomes

Example

在经过两周后，测量比较两组参与者的投资回报率，观察投资回报率是否在风险信息可视化后得到提高。

Useful idea: students want real investment consequences.

Revision point: future realized return may reflect market luck, not better risk understanding.

Q5 Better Outcomes

- Risk exposure: % allocated to risky assets
- Risk-tolerance alignment: chosen risk vs pre-measured tolerance
- Diversification: number of assets or HHI
- Ex ante Sharpe ratio: based on information available at choice
- Risk comprehension: quiz or estimated probability of loss

The primary outcome is the match between chosen portfolio risk and pre-measured risk tolerance.

Q6 Timing

Example

All outcomes will be measured immediately after participants have viewed the risk information and made their investment decision during a single experimental session lasting approximately ten minutes.

Outcomes are measured immediately after participants view the assigned risk information and submit their portfolio choice.

Immediate measurement identifies the direct treatment effect.

Q6 Timing

Example

We measured their portfolio choices one week after the information was sent.

What follow-up can show

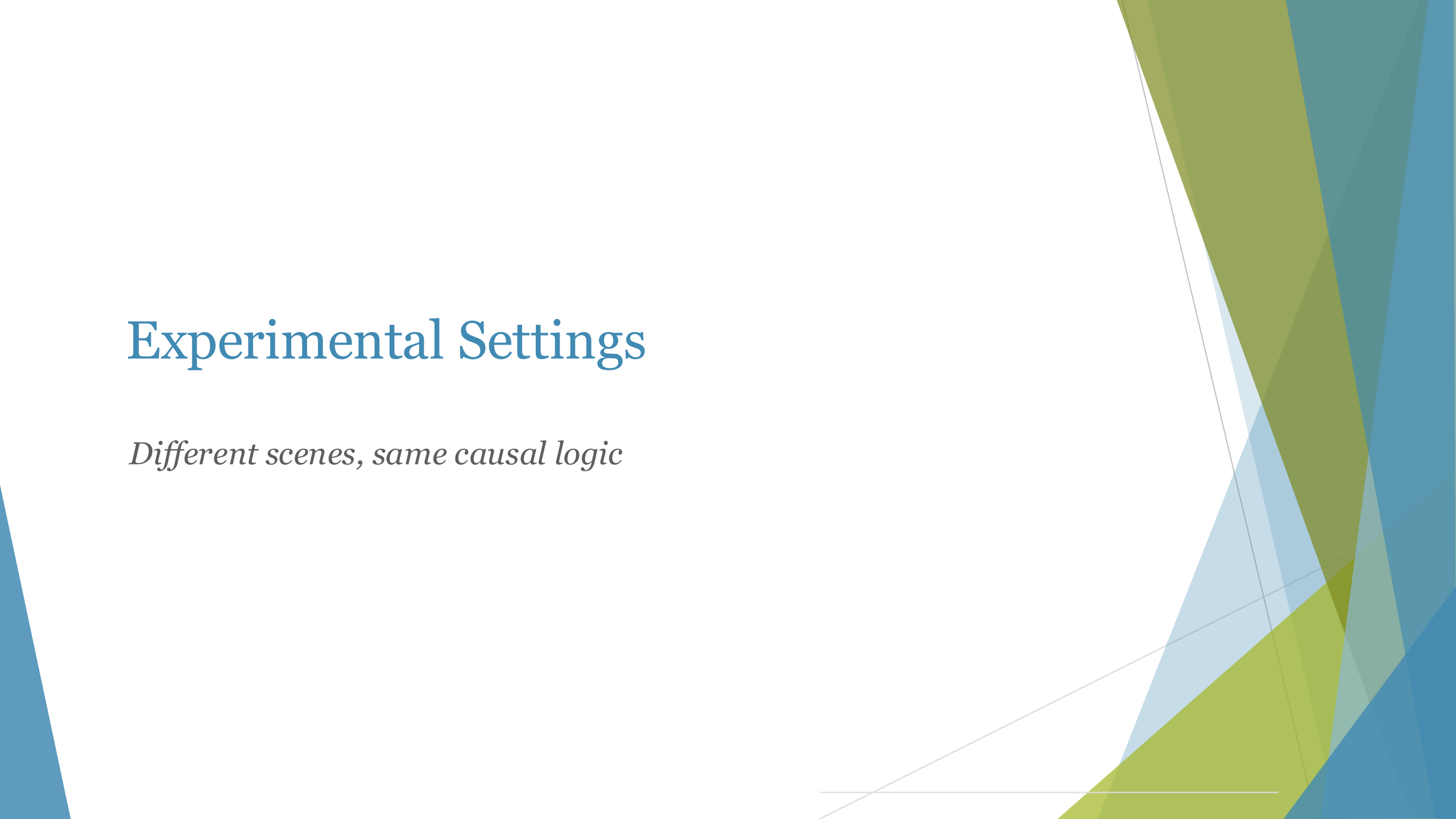
- Memory
- Persistence
- Later trading intention

What follow-up adds

- New information
- Market events
- More noise

Experimental Settings

Different scenes, same causal logic



Student Settings

Competition / simulation idea

可以采取一个“新一代最佳投资balabala比赛”的形式.....可以搭建一个模拟交易投资的系统，或者就在同花顺等软件上。

Platform A/B idea

以产品 A/B 测试的形式在后台运行.....当这些用户浏览特定高风险资产.....后端的灰度发布系统会将他们随机分为两组。

Both are useful ideas. They simply require different controls and outcome measures.

Choosing an Experimental Setting

Setting	Best for	Main risk
Online lab	Clean comparison	Lower realism
Incentivized allocation	Serious choice	Bonus design
Platform A/B test	Real behavior	Ethics/privacy
Competition	Engagement	Risk-taking pressure

The scene can differ. The causal logic still needs to be explicit.

Beyond the Six Questions

- External validity: would the result matter outside this experiment?
- Baseline information: what should be measured before treatment?
- Blinding and demand effects: can participants guess the hypothesis?

Do not add advanced terms unless they help the reader trust the design.

External Validity

- Sample: finance students or ordinary young investors?
- Task: hypothetical allocation or realistic platform behavior?
- Setting: online survey, simulation, or brokerage app?
- Time: immediate response or persistent behavior?

The more realistic the setting is, the harder it is to control.

Baseline Information

- Demographics: describe sample and check balance
- Investment experience: screen participants
- Financial literacy: balance and mechanism
- Risk preference: stratification and outcome construction
- Visual/numerical ability: who benefits from charts?
- Attention checks: exclude invalid responses

Baseline information is not decoration.

Blinding and Demand Effects

- Participants can usually tell whether they saw a chart or text.
- The bigger concern: can they guess the hypothesis?
- Use a neutral title: “A study on financial decision-making.”
- Automate assignment and material presentation.
- Anonymize group labels during analysis when possible.

Do not casually write “double-blind” unless you can explain what is blinded.

Model Experiment Section

- Participants: young investors aged 18–30 with limited experience.
- Random assignment: assigned at the individual level.
- Treatment: changes only the way risk information is presented.
- Control: same underlying risk information in text.
- Outcome: match between chosen risk and risk tolerance.
- Timing: immediately after portfolio submission.

Borrow the sentence structure, not necessarily the exact design.

Revision Exercise

- Choose one useful idea from your own answer.
- Write the clean comparison in one sentence.
- Name one primary outcome.
- Say when the outcome is measured.
- Add one sentence about external validity or demand effects.

Revision is not rejection. It is how a good intuition becomes a credible design.

A credible experiment requires more than random assignment.

We also need to justify the participants, the task, the outcome, the timing, and whether the result can generalize.

Clear design first. Sophistication second.